MTH3028 Statistical Inference

Formative coursework

Submission deadline: noon on Monday 2 November 2026

The marks for this assessment do not contribute to your final mark for the module.

Instructions

Submit your answers on ELE by the date shown above. You should submit a Zip archive containing two files: one R script (ending .R or .r) containing the R code that you used to produce your results, and one pdf file containing your report. I shall also accept report files in docx format. Your R code should be formatted to aid readability, should include brief comments to explain what each section of code does, and should run without errors. Your report (excluding your code) should comprise at most two sides of word-processed text (e.g. written with Word or LaTeX) followed by as many pages of figures or tables that you feel is appropriate. No credit will be awarded to additional pages of text. (You do not need to write two sides of text to achieve full marks—the model solution has only one side of text and about 300 words.) References, figures and tables do not count towards the two sides of text. Reference all sources that inform your work for this assignment and follow our guidance on referencing. The page size must be A4, font size at least 11 points, lines single-spaced and all margins must be at least one inch wide.

This assignment is AI-assisted. Please refer to our guidance on the use of generative artificial intelligence (GenAI) tools and to the notes on GenAI tools at the end of this document.

You may complete this assessment individually or in a group. If you work as a group then all students in the group should submit identical files and the report should include the six-digit candidate numbers of everyone in the group. Please also refer to our regulations on academic conduct and practice.

Assignment tasks

Consider modelling data $(x_1, \ldots, x_n)$ as a simple random sample from a geometric distribution with probability mass function $f(x; \phi) = \phi(1-\phi)^x$ for $x = 0, 1, 2, \ldots$, where ϕ is the parameter. This distribution has mean $\theta = (1-\phi)/\phi$ and variance $(1-\phi)/\phi^2$. You can simulate random numbers from this distribution using the R command `rgeom`.

One estimator for ϕ is $\hat{\phi} = (n-1)/(n-1+\sum_{i=1}^{n} x_i)$. From the methods discussed in lectures, create two more estimators for ϕ which differ from each other and from $\hat{\phi}$. Conduct a simulation study to compare the sampling properties of these three estimators.

Marking criteria

The maximum number of marks available is 50. Approximately 10 marks are awarded for the design of your simulation study, 10 for the implementation of your simulation study, 10 for the presentation of your results, 10 for the interpretation of your results, and 10 for the presentation of your report and code. To achieve a pass mark, you will have selected some estimators and properties to investigate, computed these for some simulated data without major errors, presented your results intelligibly, interpreted your results without major errors, and written an intelligible report. To achieve a first-class mark, you will have selected some appropriate estimators and properties to investigate, computed these accurately and precisely for an informative range of simulated data sets, presented your results effectively, interpreted your results accurately and with an appropriate level of detail, and written a clear report.

Learning outcomes

This assignment assesses the following Intended Learning Outcomes, as listed in the module descriptor: 1 to 6.

Skills developed

This assignment helps you to develop the following Skills to Thrive: confident communicator, innovative problem solver, digitally fluent, critical thinker, mathematical thinking.

Statement on use of GenAI tools

The University of Exeter is committed to the ethical and responsible use of Generative AI (GenAI) tools in teaching and learning, in line with our academic integrity policies. Direct copying of AI-generated content can be an academic offence and is addressed under plagiarism, misrepresentation and contract cheating in chapter 2 of the Academic Conduct and Practice Handbook.

This assessment is AI-Assisted. This is because using AI tools in specific ways can assist with your learning and will not inhibit fair assessment of your achievement of the module learning outcomes. It is not necessary to use AI tools, though, and you can achieve full marks without using them. However, you may use GenAI tools ethically and responsibly in the following ways to assist in the development of this assessment.

- To help you to understand key concepts and theories.
- To help you to plan the structure of your report.
- To identify errors in your computer code.
- To identify errors in your spelling or grammar.
- To format citations or references.

When writing your assessment, you must never use AI tools in the following ways.

1. For uses other than those presented in the list above.

2. To translate more than a word or short phrase into English unless agreed above.
3. To upload sensitive or identifying material to an AI tool.
4. To present material that has been generated by AI as your own work or the work of someone else.

When submitting your assessment, you must do the following.

1. Check the box during the submission process that confirms you have adhered to the university's academic conduct policy and the expectations on use of GenAI in your assessment brief.
2. Treat the AI tool like a citation from any other source.
3. Include a list of all AI prompts and hyperlinks to their output with your references, at the end of your work. You do not need to include the outputs themselves, just the links.
4. Retain the full outputs generated by the prompts you have used during your assignment. These outputs should be accessible at the hyperlinks which you have submitted with your assignment. You may be asked to produce this material in the event of an academic conduct inquiry.